Theoretical modelling in ecology and evolution

6th of May

Tom Keaney

Part 2

What is a scientific review?

Part 1

What is this and why does it matter?

Review

Cell PRESS

Intralocus sexual conflict

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Intralocus sexual conflict occurs when selection on a shared trait in one sex displaces the other sex from its phenotypic optimum. It arises because many shared traits have a common genetic basis but undergo contrasting selection in the sexes. A recent surge of interest in this evolutionary tug of war has yielded evidence of such conflicts in laboratory and natural populations. Here we highlight outstanding questions about the causes and consequences of intralocus sexual conflict at the genomic level, and its long-term implications for sexual coevolution. Whereas recent thinking has focussed on the role of intralocus sexual conflict as a brake on sexual coevolution, we urge a broader appraisal that also takes account of its potential to drive adaptive evolution and speciation.

major gaps in our understanding of IASC, clarify its potential importance in evolution and flag promising avenues of investigation.

The nature and causes of intralocus sexual conflict

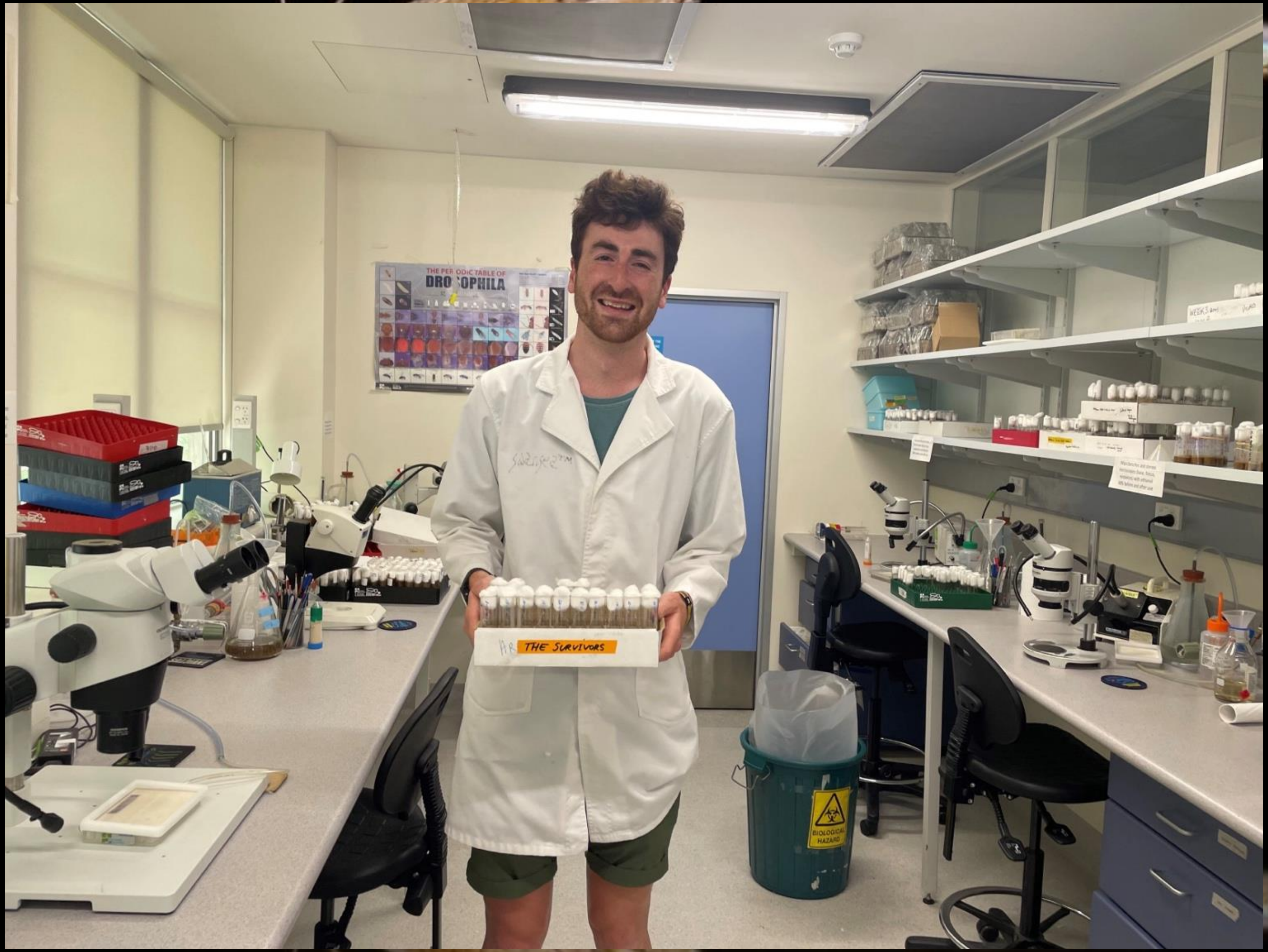
IASC reflects a conflict between shared and divergent aspects of the biology of the sexes. Shared traits are assumed to be controlled, primitively, by a common genetic machinery in both sexes [4]. This is reflected in a strong, positive intersexual genetic correlation (r_{mf}), which measures the extent of similarity between the additive effects of alleles when expressed in different sexes (Box 1). However, the sexes are defined by strongly divergent reproductive strategies that generate sex-specific selection on many shared traits, favouring the evolution of sexual

A quick detour: my path to theory

2014-16: Bachelor of Science (Zoology)
2017-18: Master of Science
2019-2023: PhD – The evolutionary consequences of separate sexes
2023-present: postdoc in Mainz

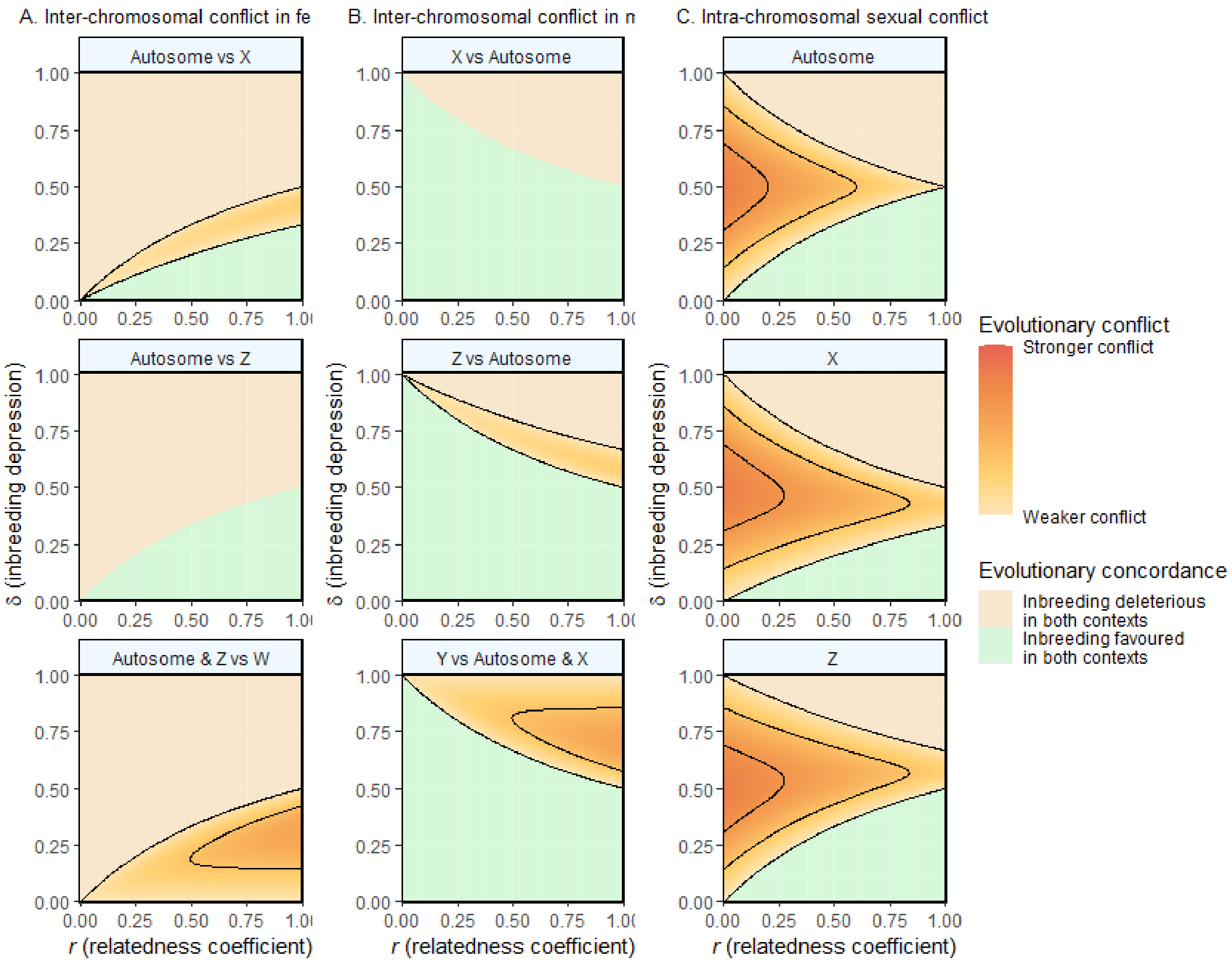
I wanted to study the ecology of wild populations…

I found I was motivated by evolutionary questions…



Questions can be formulated and logic clarified with equations

$$\delta_{\mathrm{F}} = \frac{r}{1+r}$$

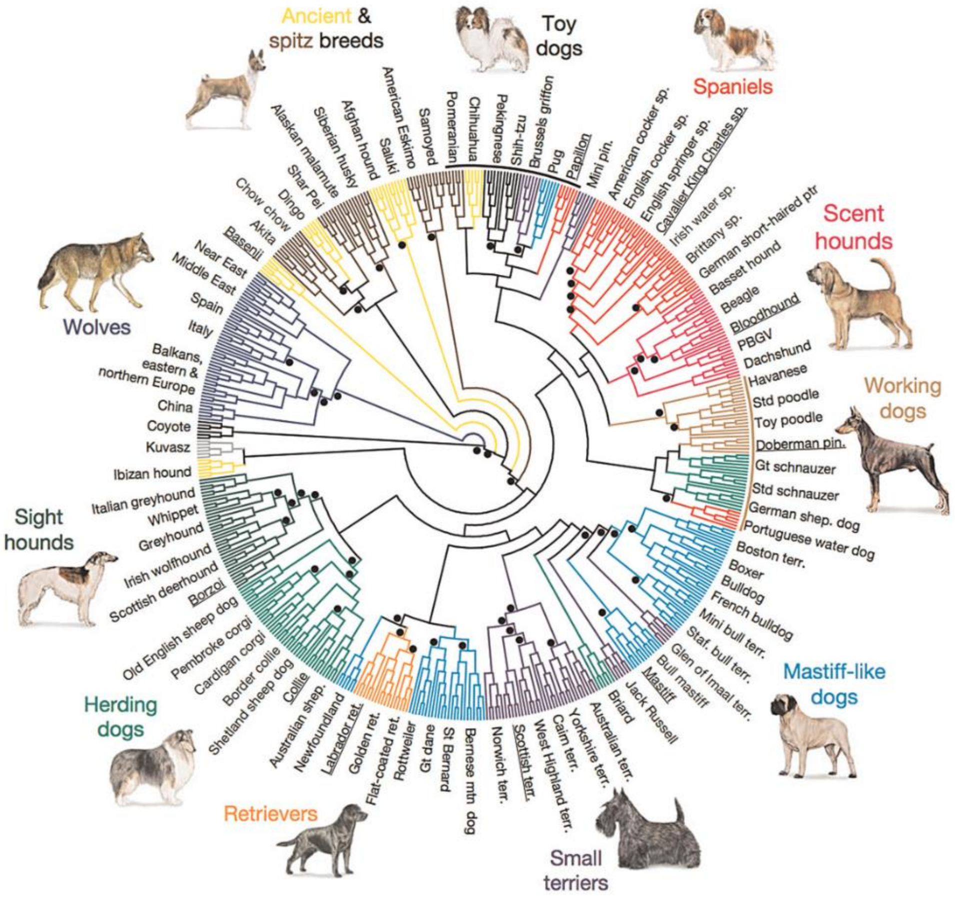


Part 1: what is intralocus conflict and why does it matter?

Some terminology

- Gene: a sequence of DNA that is inherited as a single unit over evolutionary time
- Locus: the physical position on a chromosome where a gene is found
- Allele: one variant of a gene – minor allele and major allele
- Mutation: used interchangeably with allele, particularly with minor allele
- SNP: single nucleotide polymorphism – a single position in the genome where different nucleotide bases are found within a population

Why is there so much genetic variation in our world?



vonHoldt et al. *Nature*, 2010

What is genetic variation?

All other sources of variation in the population

Phenotypic variation in the population

$$V_P = V_G + V_E$$

Genetic variation in the population
(the part important for evolution!)

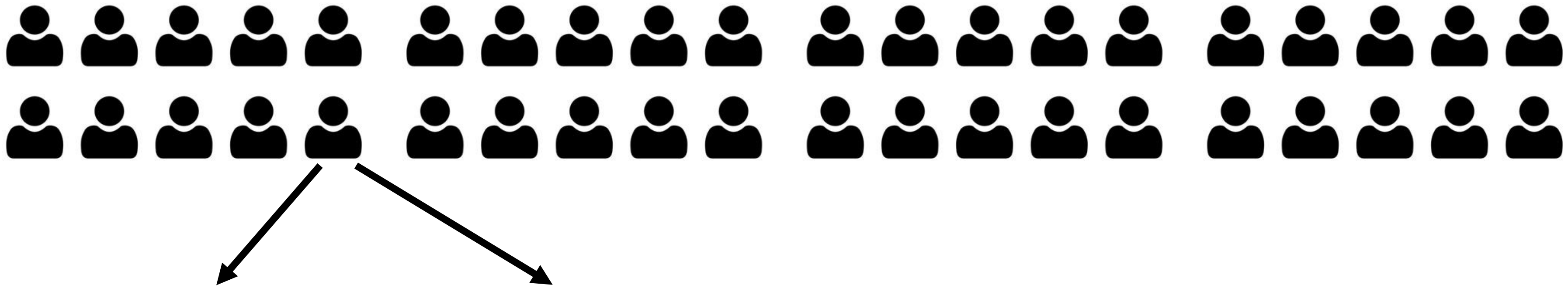
When I talk about variation, this is at the population level

Human height

~50% of the phenotypic variance in human height (European ancestry) is attributed to genetic variance

Question: is this caused by a few mutations with large effects or by many mutations with small effects?

Human height – Genome wide association studies



167cm

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Human height – Genome wide association studies

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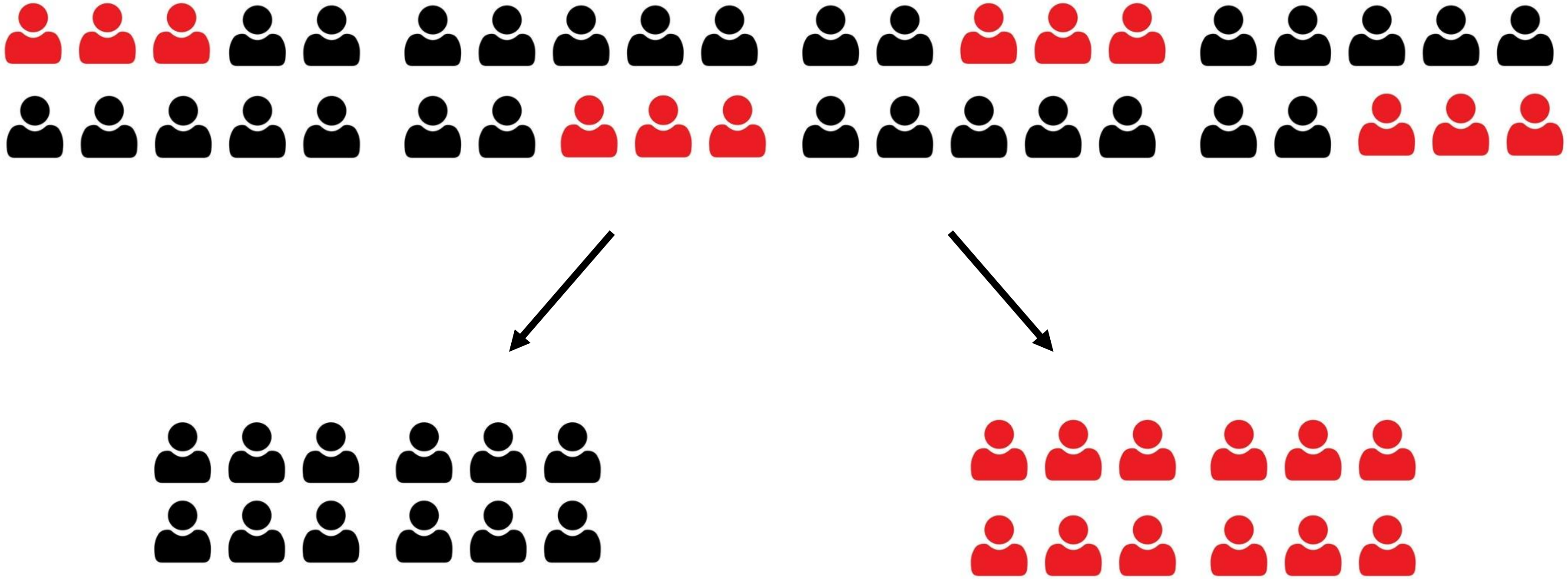
167cm

Human height – Genome wide association studies



Does the group of individuals carrying the mutation differ in height from the group that does not?

Human height – Genome wide association studies

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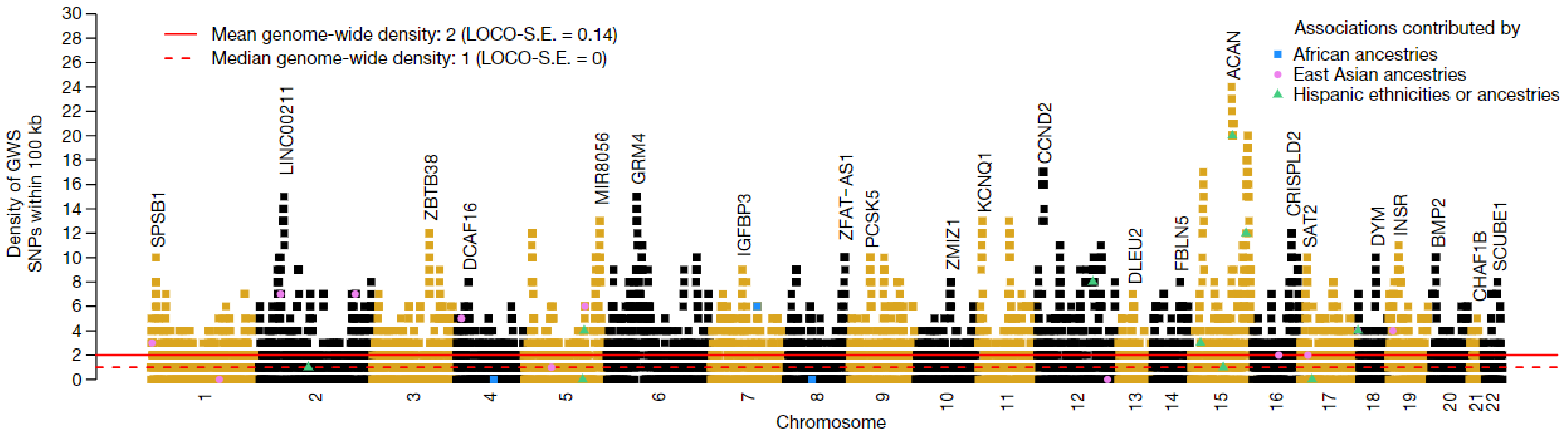
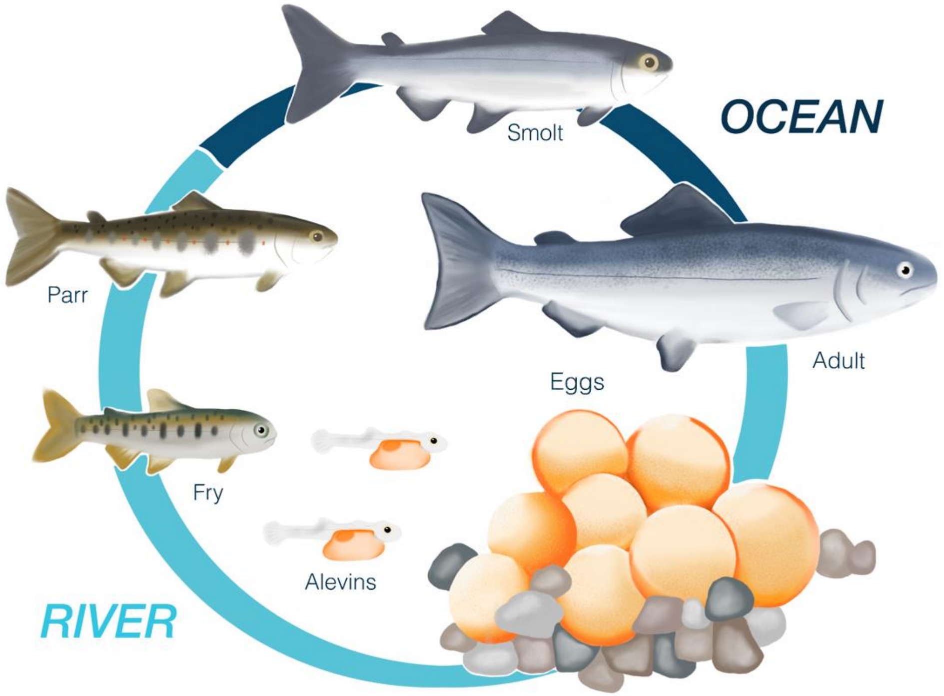
Human height

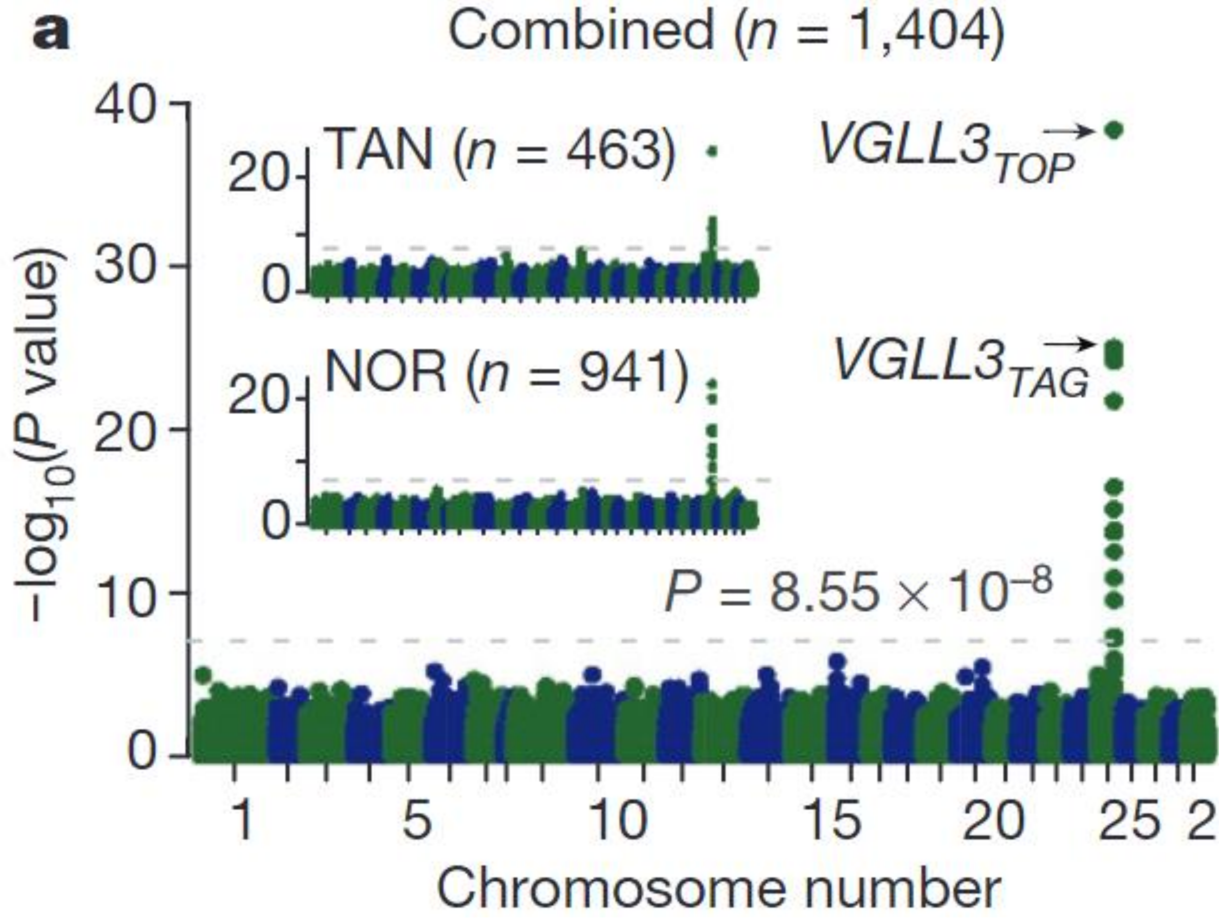
Yengo et al, *Nature* 2022



- ~50% of the phenotypic variance in human height (European ancestry) is attributed to genetic variance
- This plot shows that this genetic variation is the cumulative result of ~12,000 genetic loci!
- That means ~22% of the human genome is associated with a mutation that affects height

Some traits are not like this though…





Barson et al, *Nature* 2015

A puzzled evolutionary biologist:

“Ok, so natural selection should favour those organisms that are well-suited to their environment”

“This should result in a decrease in genetic variance, as those well-suited individuals should carry a specific, common set of alleles that convey high fitness”

“But… we find large amounts of genetic variation in traits that are important for fitness”

“What is going on?”

Mutation selection balance

- Mutations are added to the population each generation with a certain probability
- Selection subtracts mutations each generation with a certain probability
- An equilibrium frequency can be found where these two forces are balanced

Probability of mutation

$$q \approx \frac{\mu}{h(1-f)}$$

Equilibrium frequency of mutation in population

Fitness of individual carrying the mutation

Dominance

Mutation selection balance

$$q \approx \frac{\mu}{h(1-f)}$$

Probability of mutation

Equilibrium frequency of mutation in population

Dominance

Fitness of individual carrying the mutation

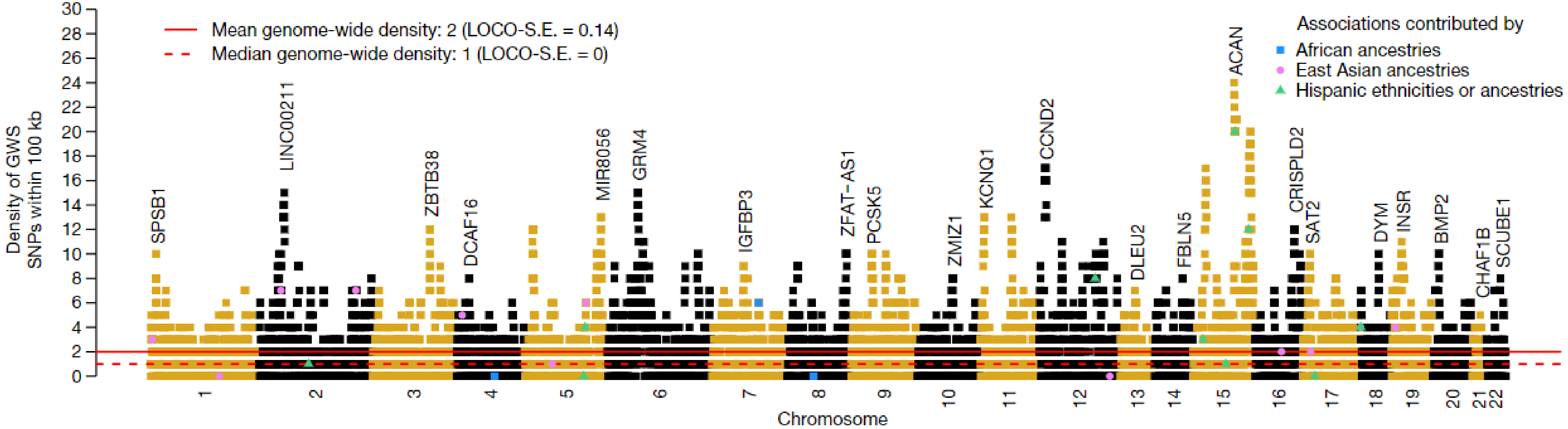
$$q \approx \frac{0.000001}{1*(1-0.99)} = 0.001$$ ~ 2 in every thousand individuals carry the mutation

$$q \approx \frac{0.000001}{1*(1-0.5)} = 0.000002$$ ~ 4 in every million individuals carry the mutation

These are results for a single locus

$$q \approx \frac{0.000001}{1*(1-0.99)} = 0.001 * 12{,}000 =$$ ~ 24 mutations in **every** individual on average

$$q \approx \frac{0.000001}{1*(1-0.5)} = 0.000002 \ * 12{,}000 =$$ ~4 in every **hundred** individuals on avg



Part 1: what is intralocus conflict and why does it matter?

Antagonistic pleiotropy

- Some genes have alleles with opposing fitness effects depending on the environment in which they are expressed.
- We call alleles with these affects antagonistically pleiotropic

Showing opposition

pleio = many
tropic = affect

- **Environment:** defined a bit weirdly here – all of the abiotic things we're familiar with + the organism the alleles inhabit, which includes the rest of the genome
- To avoid confusion, let's call this a context

Alleles are often expressed in more than one context

Whether the allele is favoured by selection depends on:

1) it’s fitness effect in each context
2) The share of generations it spends in each context

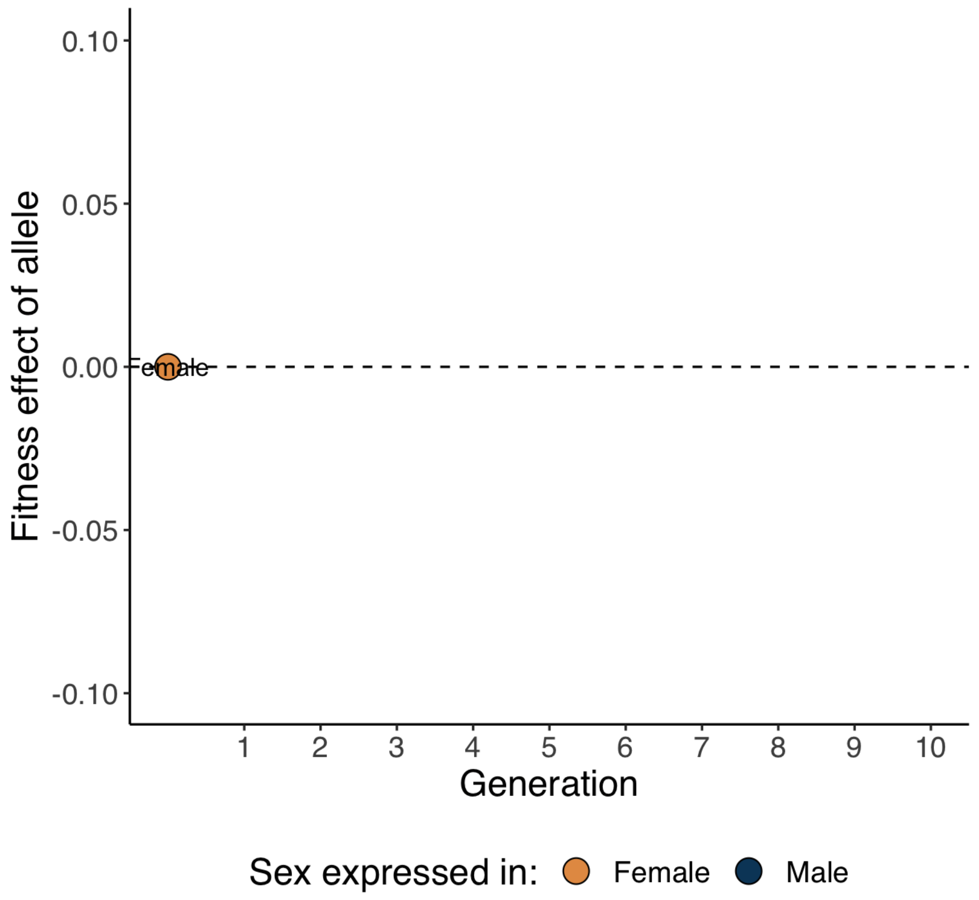
The sexes act as different contexts for alleles

Intralocus sexual conflict

	Female	Male
Big	–	+
Small	+	–

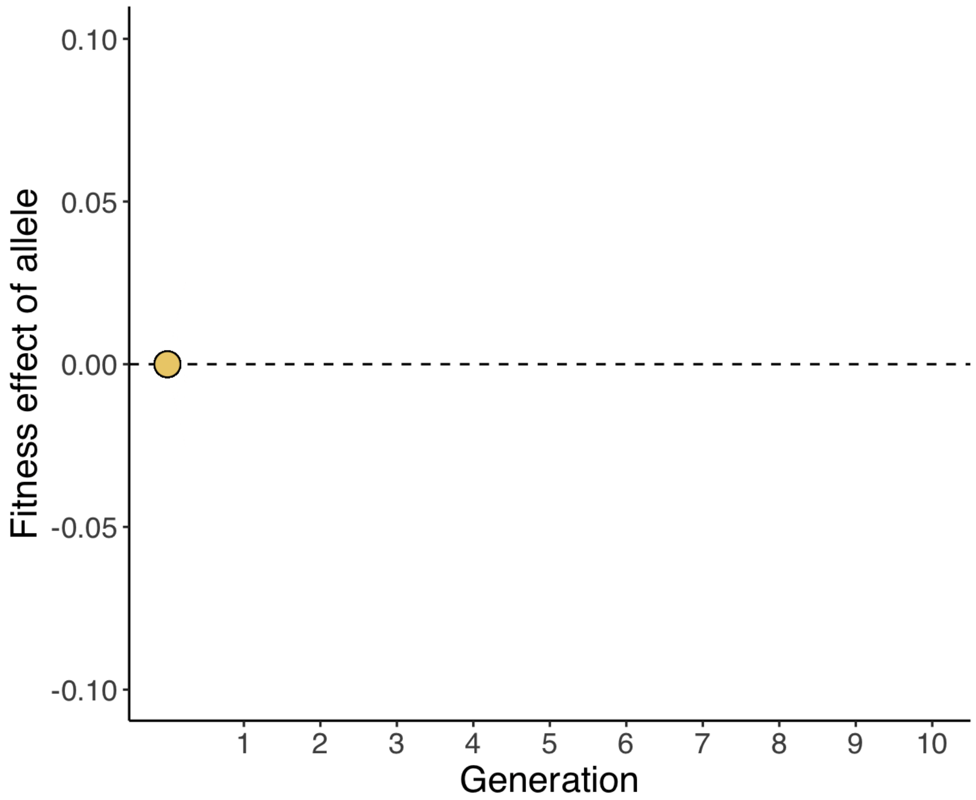
Intralocus sexual conflict

	Female	Male
Big	−	+
Small	+	−



Intralocus sexual conflict

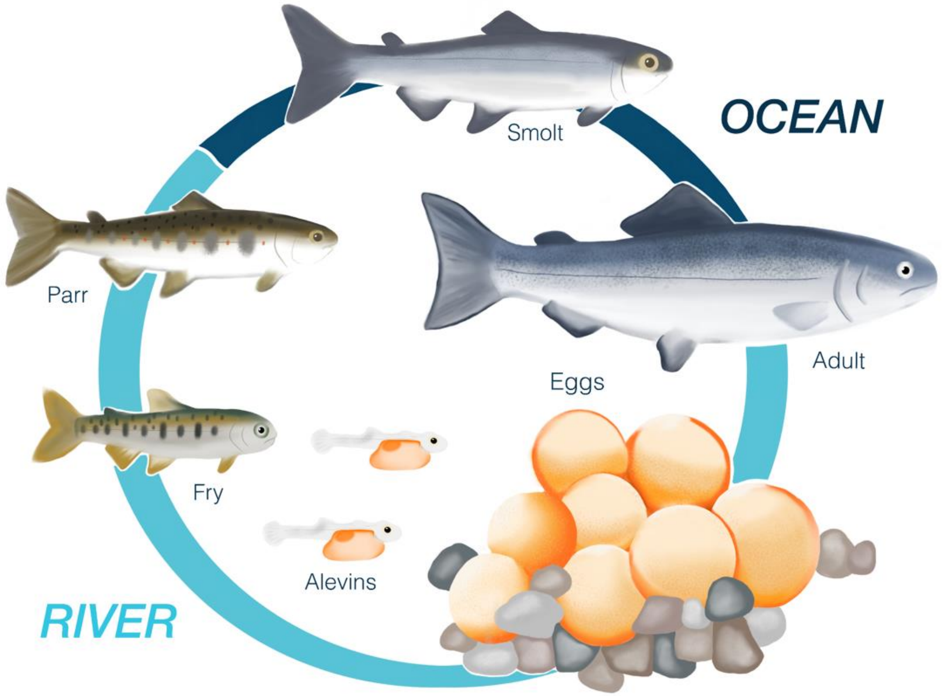
	Female	Male
Big	−	+
Small	+	−

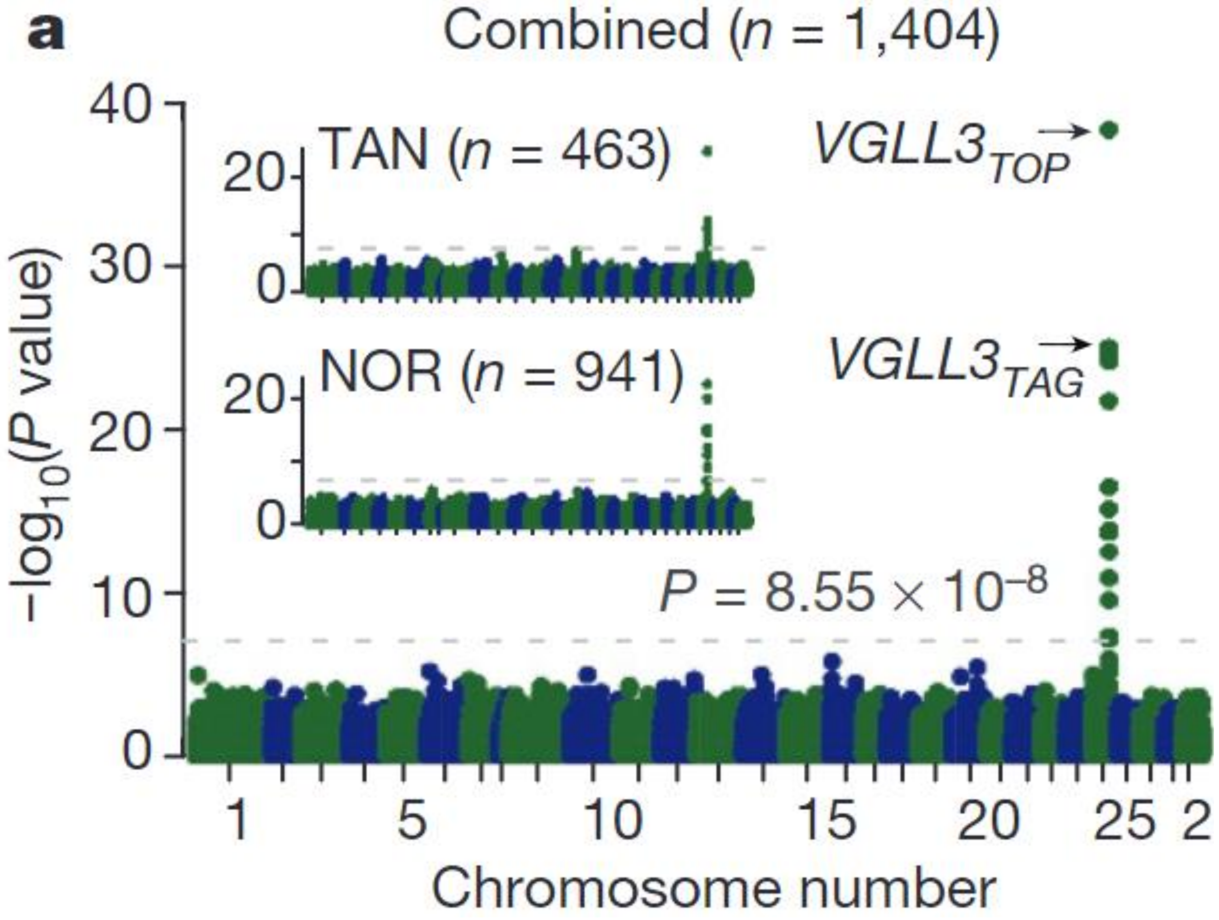


Neither allele has a consistent advantage over the other, maintaining variation at this locus

What does this mean for the sexes?

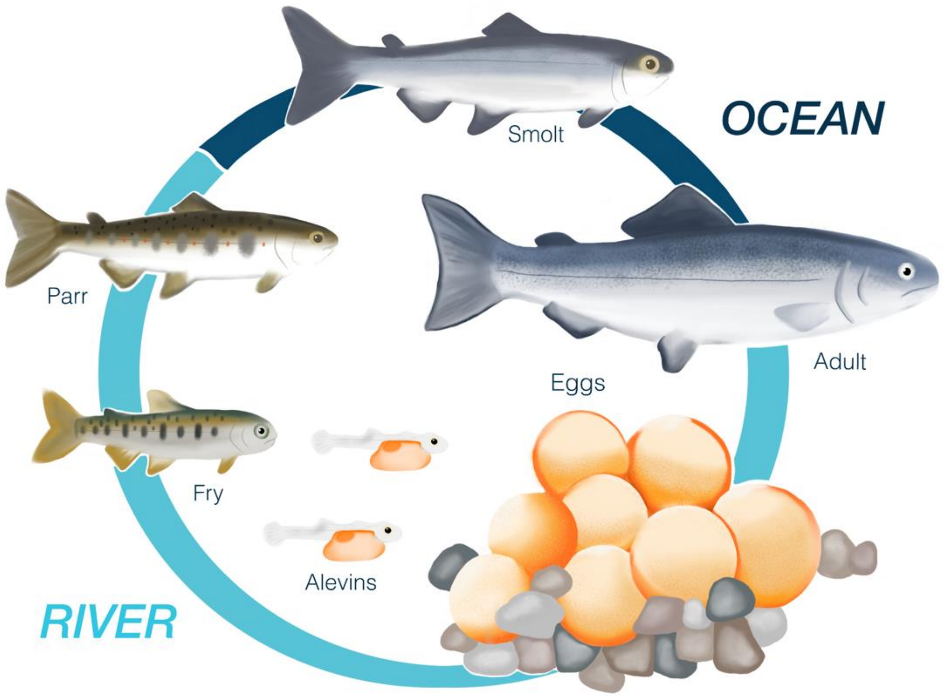
Can we explain the salmon observation?





Barson et al, *Nature* 2015

Yep, the salmon make sense

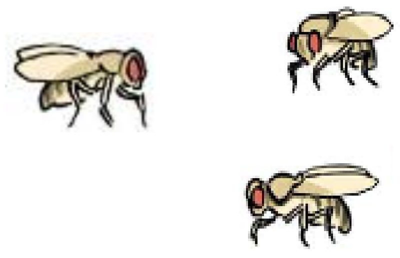


- Selection favours early maturation in males and late maturation in females
- One allele increases in frequency when in the female context, but decreases in frequency when inhabiting the male context.
- Textbook example of antagonistic pleiotropy maintaining genetic variation in a population.

Contexts can also be temporal

Consider a gene that affects thermal tolerance, with two alleles:

- The *HOT* allele increases resistance to heat stress, increasing fitness in warm conditions
- The *COLD* allele increases resistance to cold stress, increasing fitness in cold conditions



	Autumn	Spring
HOT	−	+
COLD	+	−

Selection can fluctuate with time

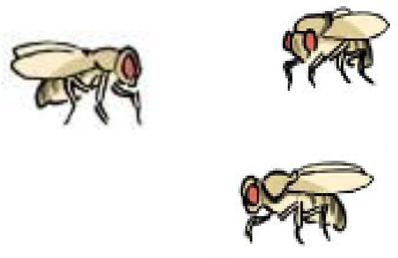
Bergland et al. *PLOS genetics*, 2014

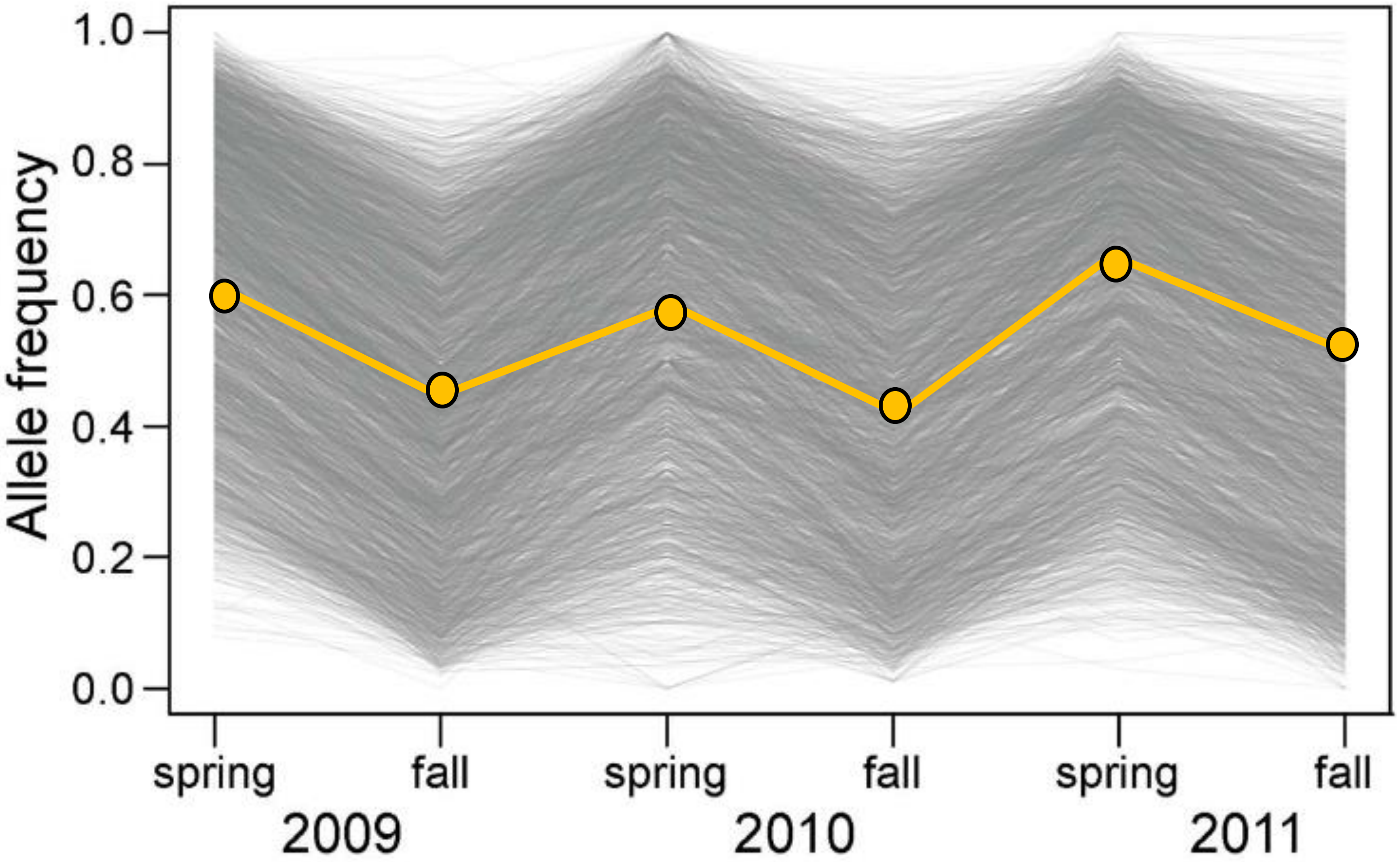
Selection can fluctuate with time

Bergland et al. *PLOS genetics*, 2014

Selection can fluctuate with time

Bergland et al. *PLOS genetics*, 2014





Different alleles are favoured in different seasons

Mutation selection balance and antagonistic pleiotropy

Mutation selection balance:

- Selection removes most mutations that have negative effects in both contexts
- Selection increases the frequency of mutations that have positive effects in both contexts

Antagonistic pleiotropy:

- Can increase the amount of standing genetic variation in a population
    - When exactly: we will model this!
- Genome should be enriched for segregating alleles that have antagonistic fitness effects

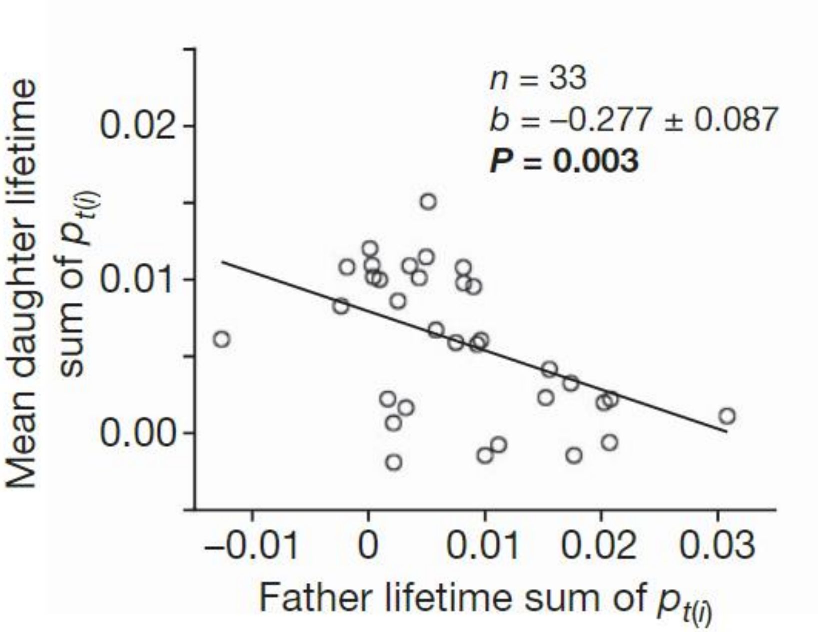
Detecting intralocus sexual conflict in real time: r_{fm}

- r_{fm}: the intersex genetic correlation for a specific trait.

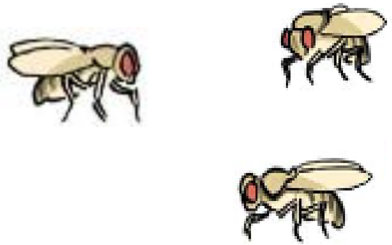
- Do alleles have similar phenotypic effects in different contexts?

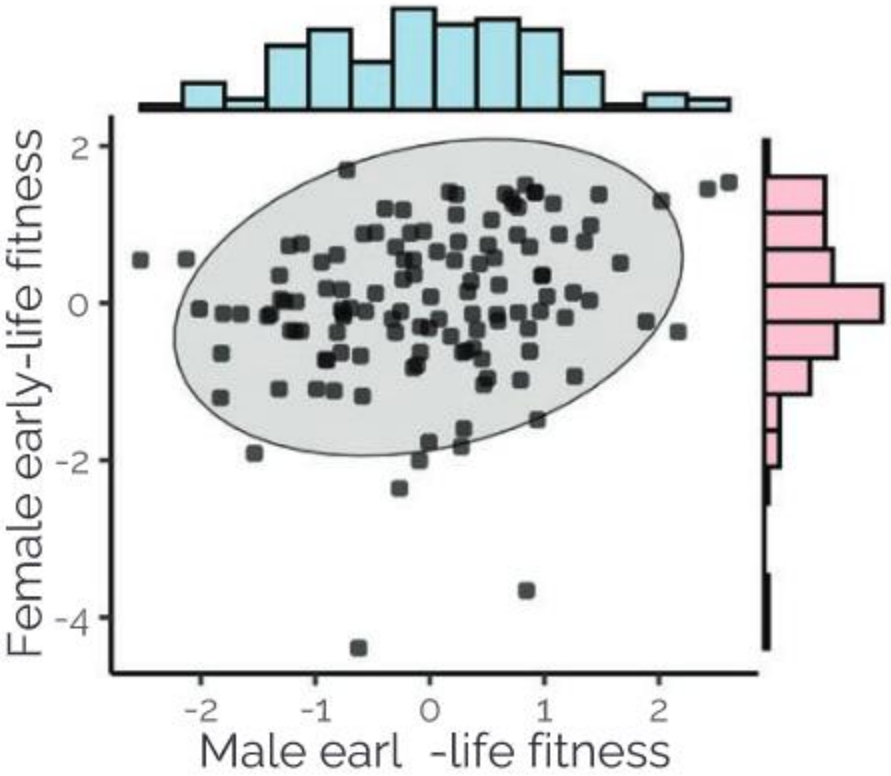
- Depends on entire genotype

- If there are strongly antagonistic alleles, or many with weaker effects, r_{fm} is predicted to be negative.



Foerster et al, *Nature* 2007





Wong and Holman, *Evolution* 2023

Important takeaways

- There’s lots of genetic variation within populations.
- Alleles inhabit a variety of different ‘contexts’.
- Optimising everything at once is rarely possible, leading to genetic compromises.
- These compromises can increase the amount of standing genetic variation in a population.

Scientific reviews

Definition of review: a comprehensive survey of the literature in order to answer a key biological question, or to identify new biological questions that need to be addressed to advance the field.

A reviewer at *Biological Reviews* described what they are looking for in a review:

"Excellent reviews provide new conceptual insight not present in the primary literature. For example, they may bring together literature items (e.g. empirical or theoretical) that were previously disconnected to show where they do in fact overlap; or the review may generate significant new ideas and hypotheses. Such reviews are rare but, if they are readable and clear, they can form the basis for a new research direction."